KILNDECK · DESIGN DECISION RECORD

KilnDeck design for construction

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DOCUMENT CONTROL

Document ID	KND-DDR-002	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kilndeck		
Source	docs/decisions/0002-design-for-construction.md @ 9f92bd8		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Design made constructable for the prototype build plan (KND-BLD-001); decided under Amish's 2026-10-03 pre-approval

0002: Design for construction

- **Date:** 2026-10-03
- **Status:** decided. Decided by Amish under his pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost."

Context

STANDARDS section 18 requires the design to be constructable before TRL 3: every part made by a stated process, every joint fixed, and an order of assembly that works (Amish, 2026-09-30: "fix the design assumptions to match and be physically feasible"). The concept of KND-PRC-001 v0.1 named eight components but no sections, joints or fixings. Writing the build plan KND-BLD-001 turned each into a part that can be cut, welded, drilled or bought, and the model `cad/src/model.py` was checked with build123d for overlaps between every pair of neighbouring parts (none left), clearances and the mass of every piece carried up the kiln.

Changes

Table 1. Changes from the concept, with the reason for each.

#	Component	The concept had	The constructable design has	Why
C1	Track	"Rails on supports" bearing on the walls	3 m modules of UPN 80 channel lying flanges down with a 20 mm round bar welded along the web; 10 mm levelling pads every 1.0 m; slotted fishplates and 6 mm gaps at joints	Stock sections, 33 kg modules two people can carry, and joints that slide 3.6 mm as the track heats by 100 K
C2	Track ends	Nothing	Bolted 200 mm end stops with rubber faces on every track end	The deck must not run off the end of a track (new R13)
C3	End trucks	Not defined	RHS 100 x 50 x 4 beam, 1.2 m wheelbase, wheel forks, seat plates for the truss feet, rubber buffers, derailment guards 24 mm either side of the track channel; boarding step on the outer truck	A lifted or broken wheel lands the truck on the guards over the track, never on the crust
C4	Wheels	"Flanged wheels"	Bought 100 mm U-groove gate wheels rated at least 500 kg. Outer-wall wheels have a 24 mm groove and guide the deck; inner-wall wheels have a 44 mm groove so the deck floats 12 mm either way	The deck grows about 4.8 mm across the trench when it warms by 60 K, and the walls are not built to a gauge tolerance
C5	Deck and guardrail	A standing platform with a separate guardrail	Two side trusses whose top chords are the top rails, each in four bolted 1,725 mm modules of 37 to 39 kg; 1.1 m top rail, knee rail and toe board	The deck must span 6.6 m; a 1.14 m deep truss does it at 11 MPa and 0.5 mm deflection, and the guardrail costs nothing extra
C6	Stiff posts	Not defined	Five frame bearers bolted to cleats on the chords with four M12 bolts each end, making U-frames at the ends and splices	A 1 kN push on the top rail at a splice puts 1.1 kN m into the post foot; the U-frame carries it into the floor
C7	Floor	"Insulated floor"	Eight bolted panels: 40 mm angle frame, 2 mm steel tread, 37 mm calcium silicate board, 1 mm aluminium underside; 30 kg each on nine bearers	Light enough to carry; the reflective underside cuts the heat from the crust
C8	Deck ends	Open	Inner end rail; outer end gate on self-closing spring hinges that opens only inward	The fireman boards from the outer wall top; the gate cannot be left open or open over the drop
C9	Moving the deck and the parking brake	Hand-pushed deck; a separate brake	Hand wheel at the outer end driving a 25 mm line shaft under the floor, with a chain drive to one wheel at each end; a spring lock pin in the hand wheel rim is the parking brake	Both ends move together so the deck cannot skew; the lock holds both driven wheels (727 N against 484 N on a 5 % grade)
C10	Hopper mounting	Hopper on the deck	Hopper carriage on its own rail along the outside of truss B, with top wheels and guide rollers on the chord, and a screw clamp	Keeps the floor whole and lets one hopper serve every hole in the row
C11	Hopper	A hopper		

#	Component	The concept had	The constructable design has	Why
			52 L, 2 mm sheet, with a bar grille (44 mm gaps) across the top	A hand cannot reach the rotor; lumps over about 44 mm stay out
C12	Metering rotor	A rotor turned by hand at the hopper	Four-pocket rotor in a welded housing; chain up to a crank over the top rail on the deck side; chain guard	The fireman turns it without leaning over the guardrail
C13	Spout	A swing spout	100 mm tube on a turned collar that swivels under the housing; outlet 1,225 mm from the deck centre line, 150 mm above the wall top	A plain swivel that can be turned on a small lathe
C14	Hot handrails	"Insulated grips"	Silicone sleeves rated to 200 °C on both top rails between the posts	A bought part that fits a 40 mm square rail
C15	Assembly	Not worked out	The deck is assembled on trestles over the cold zone of the kiln (where bricks are set or drawn, with no fire beneath) and rolled to the fire only when complete	No one works over the fire during assembly

Related choices made at the same time

#	Item	Decision	What would change it
A1	Wheel rating	At least 500 kg per wheel; the worst wheel load is about 447 kg with a 1.25 dynamic factor	A lighter deck at TRL 4
A2	Wall bearing limit until the survey	Interim limit of 0.20 MPa under a pad (conservative for an old brick wall in mud or lime mortar); the design gives 0.15 MPa	A structural survey of the partner kiln setting a measured limit
A3	Shade over the deck	No canopy on the first prototype. R7 is not met in full sun (floor top about 74 °C at 45 °C air) and is reported as such; the fireman keeps heat-resistant footwear	A wind-load check of a canopy on the rolling deck and floor temperatures logged at TRL 4
A4	Curved ends of the kiln	Not served by the prototype; R1 is met on the straight sides only	A later curved or turntable track design

Consequences

- `cad/src/model.py`, the drawing KND-DWG-001 (Rev P1), the STEP and STL files, the concept media and the BOM are generated from the constructable design. `design_state: constructable` is set in `project.yaml`.
- The parts cost rises from no estimate at TRL 1 to USD 2,627, USD 873 under the USD 3,500 value-engineering target.
- KND-CAL-001 was run on the constructable model; every requirement except R7 (full sun) and R1 (curved ends) is met on paper, R8 is met against the interim limit only, and R5 can only be checked at TRL 4.
- No change alters the pitch. The safety case is stronger, not weaker: higher guardrail, inward gate, end stops, derailment guards, a positive lock and a guarded rotor.